

MINGJUN WANG

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EDUCATION

The Chinese University of Hong Kong

2025 – Present

PhD student in COMPUTER SCIENCE & ENGINEERING

- Advisor: Prof. Bei Yu, Department of Computer Science and Engineering, CUHK

University of Chinese Academy of Sciences & ICT, CAS

2022 – 2025

Master's in Electronic Information

GPA: 3.78/4.0 (Rank: Top 5%)

- Advisor: Prof. Xiaowei Li, State Key Laboratory of Processors, Institute of Computing Technology, CAS
- Selected Courses: Deep Learning (92), Pattern Recognition (90), VLSI Testing (90)

Peking University

2018 – 2022

Bachelor's in Microelectronics Science and Technology

- Selected Courses: Neural Computation (94), Intelligent Computing Systems (100)

RESEARCH INTEREST

I am broadly interested in circuit representation learning and artificial intelligence for electronic design automation (AI4EDA) and 3D IC designs, focusing on developing novel multi-modal circuit understanding systems and neural architectures for EDA problems. Meanwhile, I have solid experience in fault simulation acceleration and analysis for circuit verification. My research spans cutting-edge AI applications in 3D IC EDA and fundamental circuit verification methodologies.

PUBLICATIONS

Conference Papers

- *Bridging Layout and RTL: Knowledge Distillation based Timing Prediction*
M. Wang, Y. Wen, B. Sun, J. Mu, J. Li, X. Wang, J. Ye, B. Yu, H. Li
International Conference on Machine Learning (ICML), 2025 **(Spotlight)**
- *CircuitNet 3.0: A Multi-Modal Dataset with Task-Oriented Augmentation for AI-Driven Circuit Design*
M. Wang, Y. Wen, Y. Lu, F. Liu, Y. Zhao, B. Han, J. Mu, Y. Lin, R. Wang, H. Li, B. Yu
The Fourteenth International Conference on Learning Representations (ICLR), 2026
- *MOSS: Multi-Modal Representation Learning on Sequential Circuits*
M. Wang, B. Sun, J. Mu, F. Gu, B. Han, T. Yang, X. Zhang, S. Liu, Y. Wen, H. Wang, J. Gao, Z. Chao, H. Han, Z. Liu, S. Liang, J. Ye, B. Yu, X. Li, H. Li
Design Automation Conference (DAC), 2025
- *EPICS: Efficient Parallel Pattern Fault Simulation for Sequential Circuits via Strongly Connected Components*
M. Wang, H. Wang, J. Mu, X. Zhang, B. Sun, Y. Wen, Z. Liu, F. Gu, J. Gao, S. Liang, J. Ye, X. Li, H. Li
Design Automation Conference (DAC), 2025
- *ETPG: Efficient Transition Fault Simulation via Dual-Strategy Pattern Parallelism and Gate Restructuring*
M. Wang, H. Wang, Z. Liu, F. Gu, J. Mu, J. Tang, J. Gao, H. Li, J. Ye, X. Li
Asia and South Pacific Design Automation Conference (ASP-DAC), 2025 **(Best Paper Nomination)**
- *Efficient Functional Safety Method for Gate-Level Fine-Grained Digital Circuits with ISO-26262*
M. Wang, H. Wang, J. Mu, Z. Liu, J. Gao, J. Ye, H. Li, X. Li
International Test Conference in Asia (ITC-Asia), 2024
- *ATLAS: Asynchronous Topological Learning for Accurate FIT Prediction via Decoupled Graph Neural Networks*
Y. Liu[†], **M. Wang**[†], S. Pei, Y. Lu, Z. Liu, B. Han, H. Li, S. Wang
Design Automation Conference (DAC), 2026
- *AnalogVerifier: A Neuro-Symbolic Framework for Analog Circuit Verification*
Y. Liu[†], **M. Wang**[†], P. Xu, R. Fu, B. Yu, T.-Y. Ho
International Conference on Machine Learning (ICML), 2026

[†] Equal contributors.

- *VIRTUAL: Vector-based Dynamic Power Estimation via Decoupled Multi-Modality Learning*
Y. Lu[†], **M. Wang**[†], Y. Wen, B. Han, J. Mu, H. Li, B. Yu
International Conference on Computer-Aided Design (ICCAD), 2025
- *DDP-Fsim: Efficient and Scalable Fault Simulation for Deterministic Patterns*
F. Gu, **M. Wang**, J. Mu, Z. Liu, J. Tang, H. Wang, Y. Wang, J. Ye, H. Li, X. Li
International Conference on Computer-Aided Design (ICCAD), 2024 **(Best Paper Nomination)**
- *DFTS: An Efficient Design-for-Test Flow for Scan Design*
J. Gao, **M. Wang**, J. Liu, W. Li, Z. Liu, J. Ye, H. Li
International Symposium of EDA (ISEDA), 2025

Journal Papers

- *DomSim: Hardware-Aware Hybrid Fault Simulation Approach with Dominator Tree-guided Partitioning*
M. Wang, H. Wang, F. Gu, Z. Liu, J. Mu, S. Liang, Z. Yu, Z. Liang, J. Gao, J. Tang, J. Ye, B. Yu, X. Li, H. Li
IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2025

Under Review

- *DistillTune: Parameter-Conditioned Distillation for Amortized Multi-Objective Bayesian Optimization in EDA*
M. Wang, Y. Liu, S. Zheng, P. Xu, Z. Lin, J. Mu, H. Li, B. Yu
IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), under review
- *Distill3D: Cross-Stage Knowledge Distillation for Early-Stage 3D IC Timing Prediction*
M. Wang, L. Zou, F. Gu, Y. Lu, B. Han, J. Mu, H. Li, B. Yu
International Conference on Computer-Aided Design (ICCAD), under review
- *ADAPT: Cross-PDK Tool Parameter Tuning for Analog Layout Synthesis*
Y. Liu[†], **M. Wang**[†], P. Xu, R. Fu, L. Jin, T.-Y. Ho
International Conference on Learning Representations (ICLR), under review
- *Graph World Models: Concepts, Taxonomy, and Future Directions*
J. Liu, S. Yang, **M. Wang**, Y. Wang, B. Yu
arXiv preprint arXiv:2604.27895, 2026

PATENTS

Invention Patents (First Author)

- *Hybrid Fault Simulation Method and Apparatus Based on Dominator Tree-Guided Circuit Partitioning*
M. Wang, H. Wang, F. Gu, J. Gao, J. Ye
CN Patent App. CN119272672A, Published Jan. 2025
- *Circuit Simulation Method, Apparatus, Computer Equipment, and Storage Medium*
M. Wang, H. Wang, F. Gu, J. Gao, J. Ye
CN Patent App. CN119272671A, Published Jan. 2025
- *FMEDA Calculation Method Integrating Diagnostic Analysis and Failure Rate*
M. Wang, H. Wang, F. Gu, J. Ye
CN Patent App. CN118897790A, Published Nov. 2024
- *Safety Analysis Method Simulating Diagnostic Mechanism Insertion*
M. Wang, H. Wang, F. Gu, J. Ye
CN Patent App. CN118897794A, Published Nov. 2024
- *Circuit Safety Assessment Method Oriented to Post-Fault Simulation*
M. Wang, H. Wang, F. Gu, J. Ye
CN Patent App. CN118897791A, Published Nov. 2024
- *Circuit Gate-Level Failure Rate Analysis Method and Apparatus*
M. Wang, H. Wang, F. Gu, J. Ye
CN Patent App. CN118897172A, Published Nov. 2024
- *Fault Diagnostic Mechanism Insertion Method and Apparatus*
M. Wang, H. Wang, F. Gu, J. Ye
CN Patent App. CN118897746A, Published Nov. 2024

[†] Equal contributors.

- *Fault Simulation Result Conversion Method and Apparatus*
M. Wang, H. Wang, F. Gu, J. Ye
CN Patent App. CN118898007A, Published Nov. 2024
- *Diagnostic Coverage Determination Method and Apparatus*
M. Wang, H. Wang, F. Gu, J. Ye
CN Patent App. CN118897793A, Published Nov. 2024
- *Data Preparation Method Oriented to Fault Simulation*
M. Wang, H. Wang, F. Gu, J. Ye
CN Patent App. CN118897792A, Published Nov. 2024

Invention Patents (Co-Author)

- *Universal Simulation Method for Permanent and Transient Stuck-at Faults*
F. Gu, **M. Wang**, J. Tang, H. Wang, J. Ye
CN Patent App. CN118917040A, Published Nov. 2024
- *Memory-Efficient Fault Simulation and Diagnosis Method via GPU*
F. Gu, Z. Chao, **M. Wang**, H. Wang, J. Ye
CN Patent App. CN118569170A, Published Aug. 2024

RESEARCH EXPERIENCE AND PROJECTS

Fault Simulation Research	Jul 2023 – Present
<i>State Key Laboratory of Processors, ICT, CAS</i>	
– Developed novel fault simulation frameworks achieving 3-10× speedup over commercial tools – Proposed hardware-aware optimization techniques for parallel fault simulation – Research resulted in 3 conference papers and 10 patents	
Circuit Multi-Modal Dataset and Machine Learning Applications	May 2024 – Present
<i>Research Project</i>	
– Built comprehensive multi-modal circuit dataset incorporating netlist, RTL, and specification data – Developed ML models for circuit functionality prediction and PPA estimation – Implemented automated pipeline for circuit representation extraction	
Industrial Experience - CASTEST	Sep 2021 – Present
<i>Beijing, China</i>	
– Led development of ISO26262-compliant safety analysis platforms for automotive chips – Optimized multi-core CPU fault simulation achieving 600× performance improvement – Contributed to commercial EDA tool development and deployment	
True Random Number Generator Based on RRAM	Sep 2020 – Jun 2022
<i>Undergraduate Thesis Project</i>	
– Designed and implemented RRAM-based random number generator – Achieved high randomness passing NIST statistical test suite	

AWARDS AND HONORS

International Exchange Scholarships for Empowerment, 2025 (Awarded by Institute of Computing Technology, CAS).

Director's Special Award, 2024 (1st out of 400 students, awarded by Institute of Computing Technology, CAS).

National Scholarship, 2024 (Top 2% of students nationwide).

First Prize, EDA² Integrated Circuit Elite Challenge 2023 (1st out of 230 teams, awarded by Huawei HiSilicon).

Special Prize, LLM4EDA Summer School Hackathon 2024 (1st out of 100 participants, awarded by National Key Laboratory of Integrated Chips and Systems).

Winner, IEEE ICLAD-DAC GenAI Hackathon 2025.

Second Place, EDathon 2025, Programming Competition on Electronic Design Automation (Awarded by IEEE CEDA).

Second Prize (Enterprise Track), The 8th China Graduate IC Innovation Contest 2025.

First Prize, EDA² Integrated Circuit Elite Challenge 2023 (1st out of 230 teams, awarded by Huawei HiSilicon).

Academic Endeavor Award, 2024 (Awarded by the EDA Elite Challenge for research excellence).

Huawei Intelligent Base Scholarship, 2023 (Awarded to top 5% of students by Huawei Technologies).

Cambricon Outstanding Student Award, 2023 (Awarded to 10/300 students by Cambricon Technologies).

First Prize of Academic Scholarship, Chinese Academy of Sciences, 2024 (Top 15% of students in CAS).

Outstanding Contribution Award for Beijing 2022 Winter Olympics, 2022 (Top 10% of volunteers in PKU).

Peking University Social Work Award, 2020 (Awarded for exceptional contributions to social work).

Peking University Social Work Award, 2019 (Awarded for exceptional contributions to social work).

LEADERSHIP AND VOLUNTEER WORK

Volunteer Leader, Beijing Winter Olympics 2022

Oct 2021 – Mar 2022

International Olympic Committee

- Served as Olympic Family Assistant (OFA), managing VIP reception, scheduling, and transport arrangements.
- Contributed to 500-day preparation work, including event coordination, promotional design, and volunteer management.

President, Student Science and Technology Association

Sep 2023 – Dec 2024

Institute of Computing Technology, CAS

- Organized 16 science popularization sessions covering fundamental computer science topics.
- Promoted the "Spring Equinox Science Program," boosting students' interest in technology.

Deputy Secretary-General, PKU Student Activity Center

Oct 2020 – Apr 2022

Peking University

- Organized university-wide events, including volunteer commendation galas and Winter Olympics volunteer recruitment campaigns.

SKILLS

Programming: Verilog, SystemVerilog, Python, C++, etc.

EDA Tools: Synopsys VCS, Design Compiler, Cadence Xcelium, etc.

Frameworks: PyTorch, TensorFlow

Others: Git, Linux, LaTeX